

Question

Apply extended ACL

enable

configure terminal

hostname R1

interface gig0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gig0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

interface gig0/2

ip address 192.168.3.1 255.255.255.0

no shutdown

exit

router rip

version 2

network 192.168.1.0

network 192.168.2.0

network 192.168.3.0

no auto-summary

exit

banner motd ^C

Name: John Doe | Roll Number: 2024CS001

^C

access-list 100 deny ip host 192.168.1.1 192.168.2.0 0.0.0.255

access-list 100 deny ip host 192.168.1.2 192.168.2.0 0.0.0.255

access-list 100 deny ip 192.168.3.0 0.0.0.255 192.168.2.0 0.0.0.255

access-list 100 permit ip any any

```
show access-list 100

interface gig0/0

ip access-group 100 in

exit

interface gig0/2

ip access-group 100 in

exit

write memory

show run | section access-list
```

Testing

```
# From PC 192.168.1.1

ping 192.168.2.100
```

```
# From PC 192.168.1.2

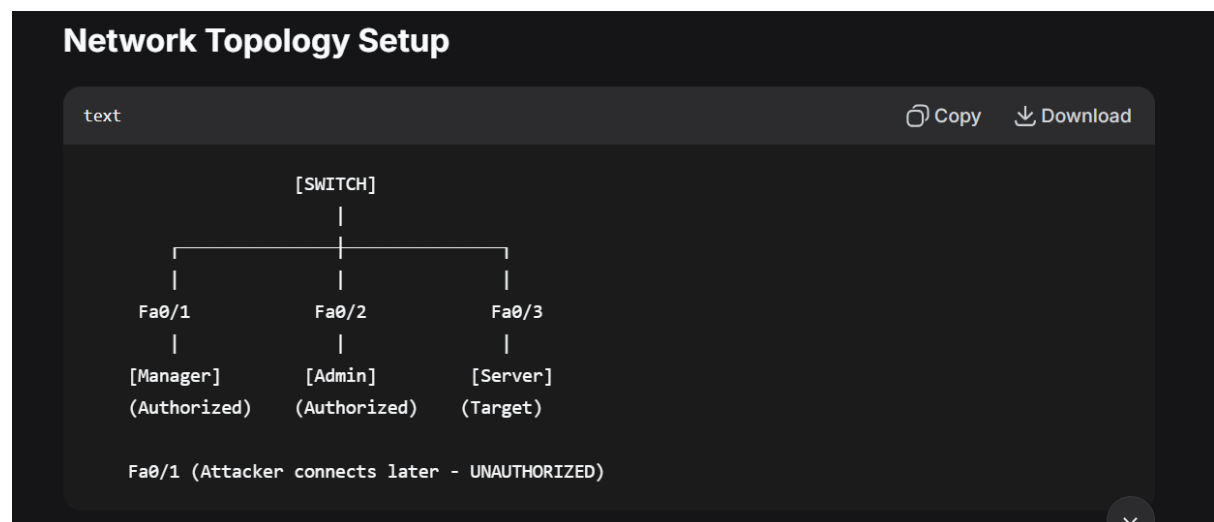
ping 192.168.2.100
```

```
# From PC 192.168.3.100

ping 192.168.2.100
```

Question

Configure Port Security



Switch Config

enable

configure terminal

hostname SW1

Set banner with Name and Roll Number

banner motd ^C

Name: [YOUR_NAME] | Roll Number: [YOUR_ROLL_NUMBER]

Config port security on manager

Navigate to interface Fa0/1

interface fastEthernet 0/1

Enable port security

switchport port-security

Set maximum MAC addresses to 1

switchport port-security maximum 1

Set violation mode to shutdown

switchport port-security violation shutdown

Allow only Manager's MAC address

Method 1: Using sticky MAC (recommended)

switchport port-security mac-address sticky

Method 2: Manual MAC entry (if you know Manager's MAC)

switchport port-security mac-address [MANAGER_MAC_ADDRESS]

Shutdown and no shutdown to apply (if port was up)

shutdown

no shutdown

exit

Config Port Security on Admin

interface fastEthernet 0/2

Enable port security

switchport port-security

Set maximum MAC addresses to 1

switchport port-security maximum 1

Set violation mode to shutdown

switchport port-security violation shutdown

Use sticky MAC learning

switchport port-security mac-address sticky

Reset port

shutdown

no shutdown

exit

Verify Configuration

Check port security on specific interfaces

show port-security interface fastEthernet 0/1

show port-security interface fastEthernet 0/2

Show all port security information

show port-security

Show running configuration for port security

show running-config | section fastEthernet

Show MAC address table

show mac address-table

Testing

Manager:

Ping the server

ping 192.168.1.100

(Assuming server IP is 192.168.1.100)

Expected: SUCCESSFUL ping replies

Switch:

show port-security address

show mac address-table interface fastEthernet 0/1

Admin:

Ping the server

ping 192.168.1.100

Expected: SUCCESSFUL ping replies